WIA1005 Network Technology Foundation

Chapter 3

Network Access and Ethernet

Contents

- Physical Layer
- Network Media
- Data Link Layer
- Media Access Control
- Ethernet
- Switching
- Address Resolution Protocol

Physical Layer

- Physical layer provides the means to transport the bits that make up a data link layer frame across the network media.
- This layer accepts a complete frame from the data link layer and encodes it as a series of signals that are transmitted onto the local media.
- The physical layer produces the representation and groupings of bits for each type of media as:
  - **Copper cable**: The signals are patterns of electrical pulses.
  - **Fiber-optic cable**: The signals are patterns of light.
  - **Wireless**: The signals are patterns of microwave transmissions.

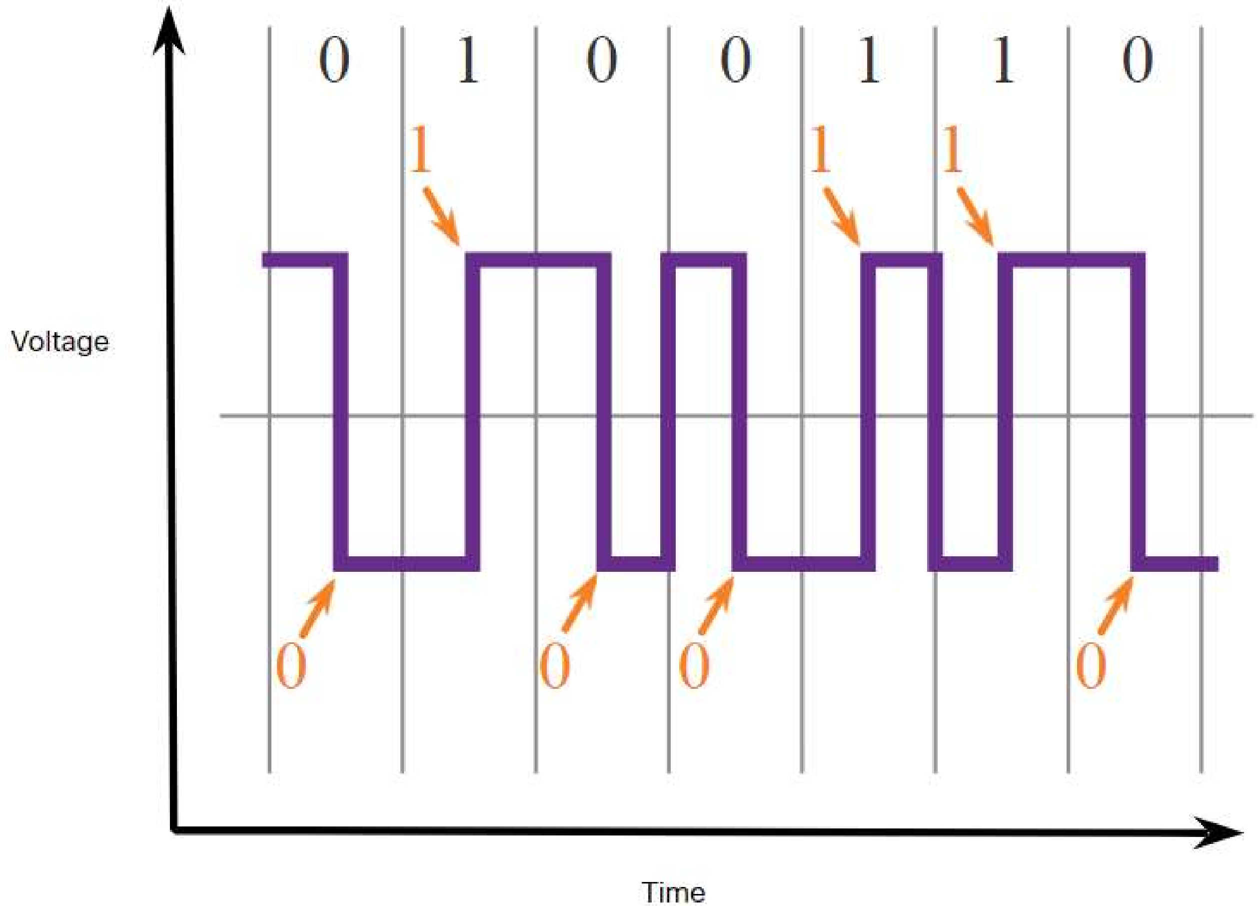
Physical Layer

- The physical layer standards address three functional areas:
- **Physical Components**
  - The physical components are the electronic hardware devices, media, and other connectors that transmit and carry the signals to represent the bits.
  - Hardware components such as network adapters (NICs), interfaces and connectors, cable materials, and cable designs are all specified in standards associated with the physical layer.

Physical Layer

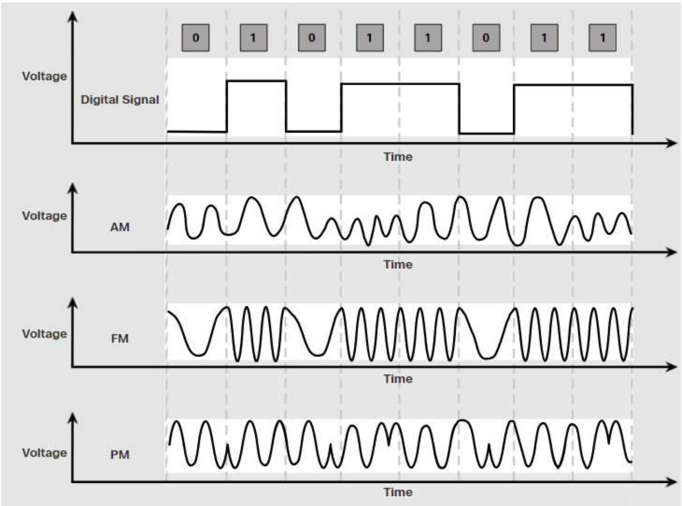
- **Encoding**
  - Encoding is a method of converting a stream of data bits into a predefined "code”. Codes are groupings of bits used to provide a predictable pattern that can be recognized by both the sender and the received.
  - The codes can be used for control purposes such as identifying the beginning and end of a frame.
  - **Manchester encoding**: A 0 is represented by a high to low voltage transition and a 1 is represented as a low to high voltage transition.
  - **Non-Return to Zero (NRZ)**: A 0 may be represented by one voltage level on the media and a 1 might be represented by a different voltage on the media.

Physical Layer



Physical Layer

- **Signaling**
  - The method of representing the bits is called the signaling method. The physical layer standards must define what type of signal represents a "1" and what type of signal represents a "0".
  - For example, a long pulse might represent a 1 whereas a short pulse might represent a 0



Physical Layer

- **Bandwidth** is the capacity of a medium to carry data. Digital bandwidth measures the amount of data that can flow from one place to another in a given amount of time.
- Bandwidth is typically measured in **kilobits per second (kbps), megabits per second (Mbps)** or or **gigabits per second (Gbps)**.
- **Throughpu**t is the measure of the transfer of bits across the media over a given period of time. Many factors influence throughput including:
  - The amount of traffic
  - The type of traffic
  - The latency created by the number of network devices encountered between source and destination

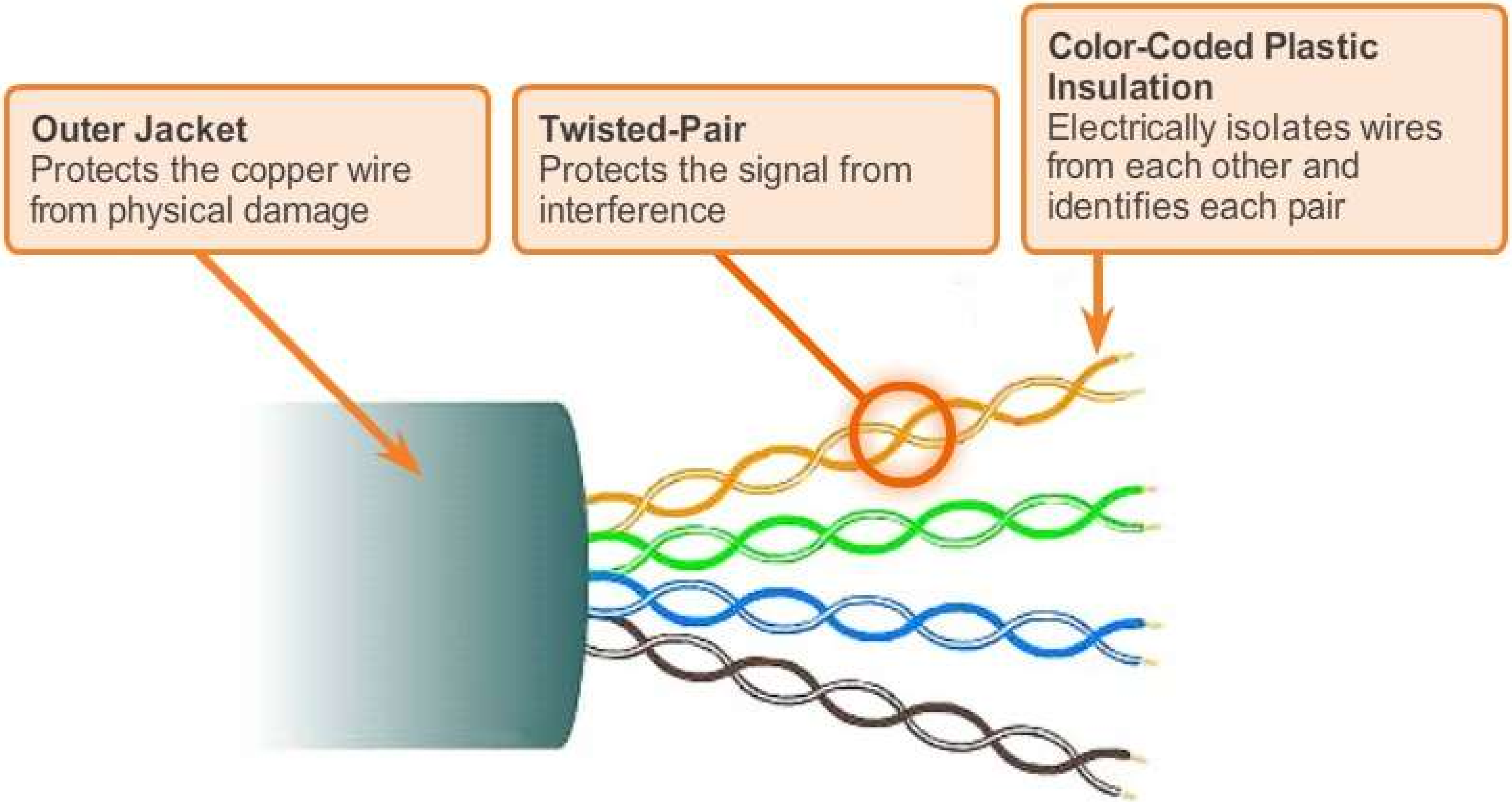
Physical Layer

- Latency refers to the amount of time, including delays, for data to travel from one given point to another.
- Goodput is the measure of **usable data transferred** over a given period of time.
- Goodput is throughput minus traffic overhead for establishing sessions, acknowledgments, encapsulation, and retransmitted bits.
- Goodput is always lower than throughput, which is generally lower than the bandwidth.

Network Media

- Networks use copper media because it is inexpensive, easy to install, and has low resistance to electrical current. However, copper media is limited by distance and signal interference.
- There are three main types of copper media used in networking:
  - **Unshielded Twisted-Pair (UTP)**
  - **Shielded Twisted-Pair (STP)**
  - **Coaxial**
- Unshielded twisted-pair (UTP) cabling is the most common networking media terminated with **RJ-45** connectors

Network Media



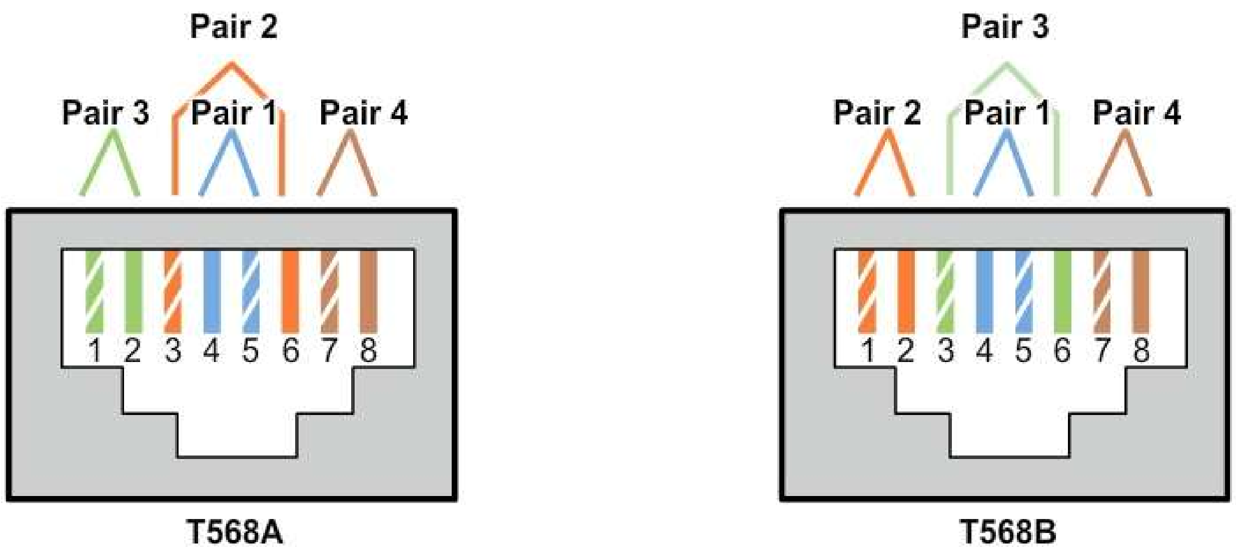
Network Media

- Shielded twisted-pair (STP) provides better noise protection than UTP cabling. However, compared to UTP cable, STP cable is significantly more expensive and difficult to install. Like UTP cable, STP uses an RJ-45 connector.
- Coaxial cables attach antennas to wireless devices. The coaxial cable carries radio frequency (RF) energy between the antennas and the radio equipment. Coaxial cable also used for **Cable Internet installations.**

Network Media

- Different situations may require UTP cables to be wired according to different wiring conventions.
- **Ethernet Straight-through**: The most common type of networking cable. It is commonly used to interconnect a **host to a switch and a switch to a router**.
- **Ethernet Crossover**: An uncommon cable used to interconnect similar devices together. For example to connect a **switch to a switch, a host to a host, or a router to a router**. Auto-MDIX cam detect cable type automatically.
- **Rollover**: A Cisco proprietary cable used to connect to a router or switch console port.

Network Media



Cable Type	Standard	Application
Ethernet Straight-through	Both ends T568A or both ends T568B	Connects a network host to a network device such as a switch or hub.
Ethernet Crossover	One end T568A, other end T568B	• Connects two network hosts • Connects two network intermediary devices (switch to switch, or router to router)
Rollover	Cisco proprietary	Connects a workstation serial port to a router console port, using an adapter.

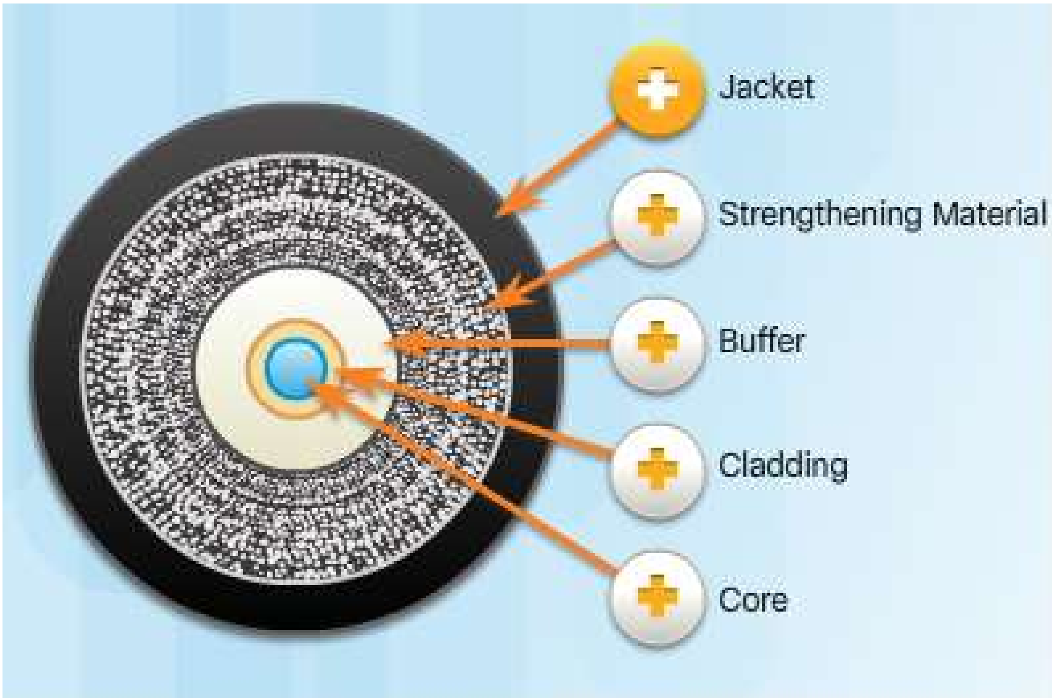
Network Media

- **Optical fiber cable** permits the transmission of data over longer distances and at higher bandwidths than any other networking media.
- Fiber-optic cabling is used in four types of industry:
  - **Enterprise Networks**
  - **FTTH and Access Networks**: Fiber-to-the-home
  - **Long-Haul Networks**: Networks typically range from a few dozen to a few thousand kilometers and use up to 10 Gbps-based systems.
  - **Submarine Networks**: Special fiber cables are used to provide reliable high-speed, high-capacity solutions capable of surviving in harsh undersea environments up to transoceanic distances.

Network Media

- Fiber-optic cables can be broadly classified into two types:
- **Single-mode fiber (SMF)**: Consists of a very small core and uses expensive **laser** technology to send a single ray of light. Popular in **long-distance** situations spanning hundreds of kilometers such as required in long haul telephony and cable TV applications.
- **Multimode fiber (MMF)**: Consists of a larger core and uses **LED** emitters to send light pulses. Specifically, light from an LED enters the multimode fiber at different angles. Popular in LANs because they can be powered by low cost LEDs. It provides bandwidth up to 10 Gb/s over link lengths of up to 550 meters.

Network Media



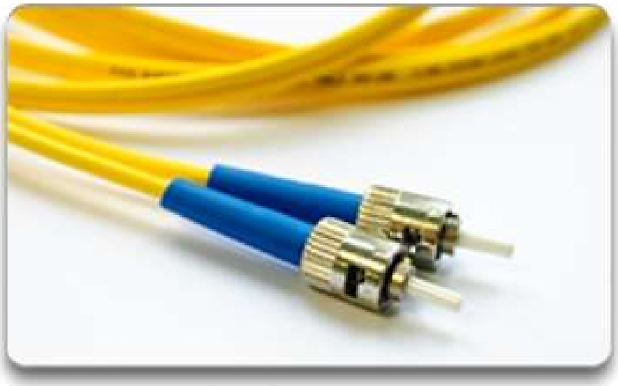
Fiber Media Cable Design

Network Media

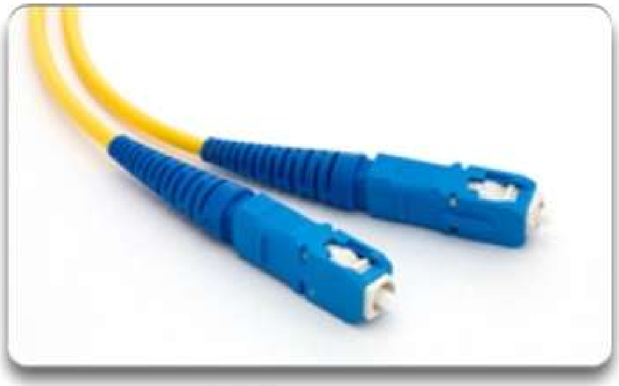
- Fiber-optic connectors include:
- **Straight-Tip (ST)**: An older bayonet style connector widely used with multimode fiber.
- **Subscriber Connector (SC)**: Sometimes referred to as square connector or standard connector. It is a widely adopted LAN and WAN connector that uses a push-pull mechanism to ensure positive insertion. This connector type is used with multimode and single-mode fiber.
- **Lucent Connector (LC)**: Sometimes called a little or local connector, is quickly growing in popularity due to its smaller size. It is used with single-mode fiber and also supports multimode fiber.

Network Media

Fiber Optic Connectors

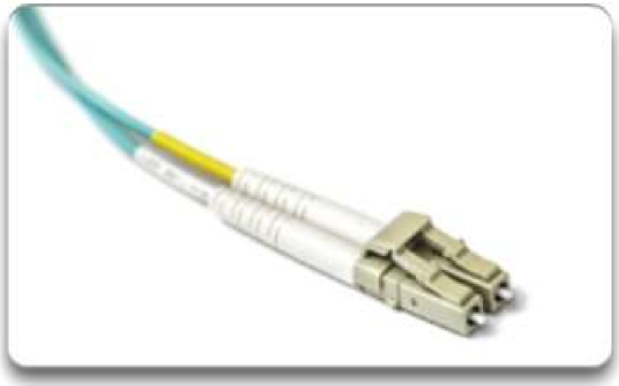


ST Connectors



SC Connectors

LC Connector



Duplex Multimode LC Connectors

Network Media

UTP and Fiber-Optic Cabling Comparison

Implementation Issues	UTP Cabling	Fiber-Optic Cabling
Bandwidth supported	10 Mb/s - 10 Gb/s	10 Mb/s - 100 Gb/s
Distance	Relatively short (1 - 100 meters)	Relatively long (1 - 100,000 meters)
Immunity to EMI and RFI	Low	High (Completely immune)
Immunity to electrical hazards	Low	High (Completely immune)
Media and connector costs	Lowest	Highest
Installation skills required	Lowest	Highest
Safety precautions	Lowest	Highest

Network Media

- Wireless media carry electromagnetic signals that represent the binary digits of data communications using radio or microwave frequencies.
- **Coverage area**: Certain construction materials used in buildings and structures, and the local terrain, will limit the effective coverage.
- **Interference**: Wireless is susceptible to interference and can be disrupted by such common devices as household cordless phones, some types of fluorescent lights, microwave ovens, and other wireless communications.
- **Security**: Network security is a major component of wireless network administration.

Network Media

- IEEE 802.11 standards
- Commonly referred to as Wi-Fi
- Uses CSMA/CA
- Variations include:
  - 802.11a: 54 Mb/s, 5 GHz
  - 802.11b: 11 Mb/s, 2.4 GHz
  - 802.11g: 54 Mb/s, 2.4 GHz
  - 802.11n: 600 Mb/s, 2.4, and 5 GHz
  - 802.11ac: 1 Gb/s, 5 GHz
  - 802.11ad: 7 Gb/s, 2.4 GHz, 5 GHz, and 60 GHz

- IEEE 802.15 standard
- Supports speeds up to 3 Mb/s
- Provides device pairing over distances from 1 to 100 meters

- IEEE 802.16 standard
- Provides speeds up to 1 Gb/s
- Uses a point-to-multipoint topology to provide wireless broadband access

Data Link Layer

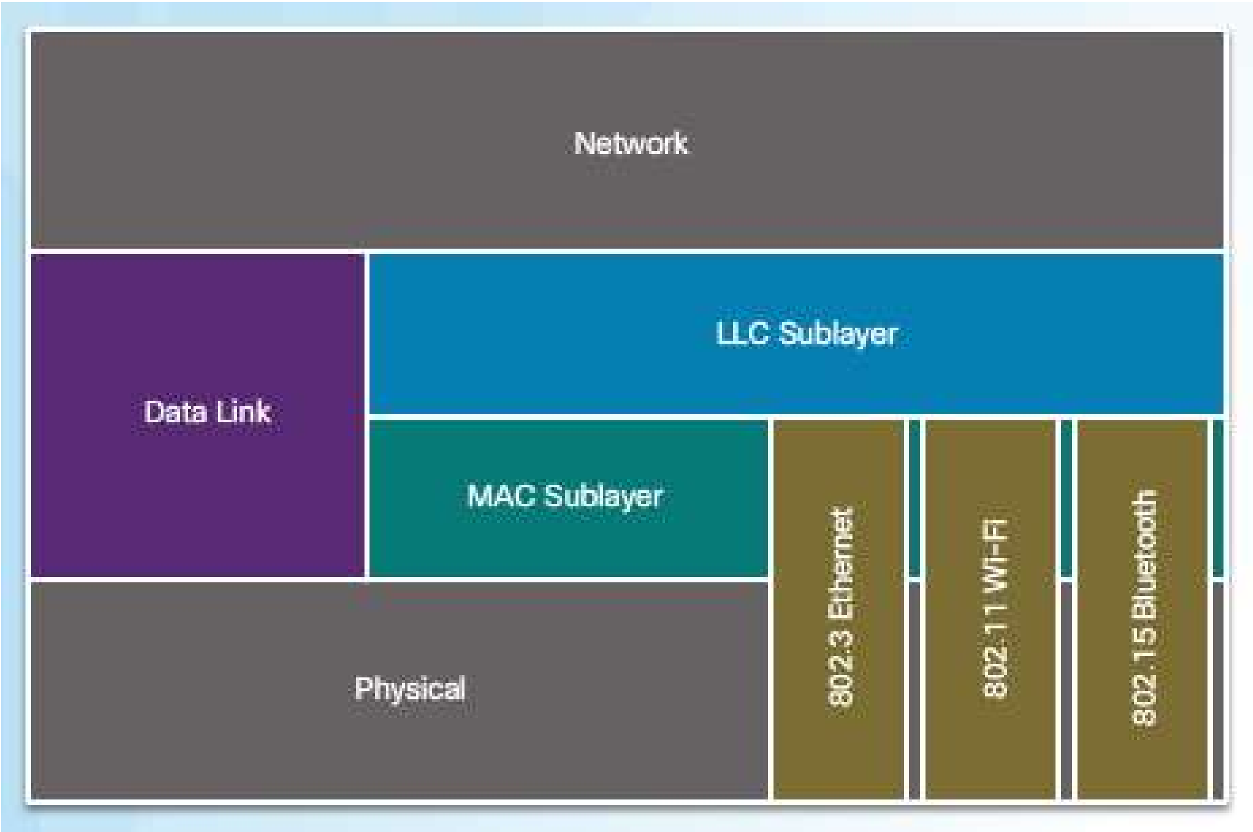
- Data link layer is responsible for the exchange of frames between nodes over a physical network media. It controls media access control and performs error detection.
- Data link layer also accepts Layer 3 packets and packages them into data units called frames.
- The data link layer is actually divided into two sublayers:
  - **Logical Link Control (LLC)**: This upper sublayer defines the software processes that provide services to the network layer protocols.

Data Link Layer

- **Media Access Control (MAC)**: This lower sublayer defines the media access processes performed by the hardware. It provides data link layer addressing and delimiting of data according to the physical signaling requirements of the medium and the type of data link layer protocol in use.

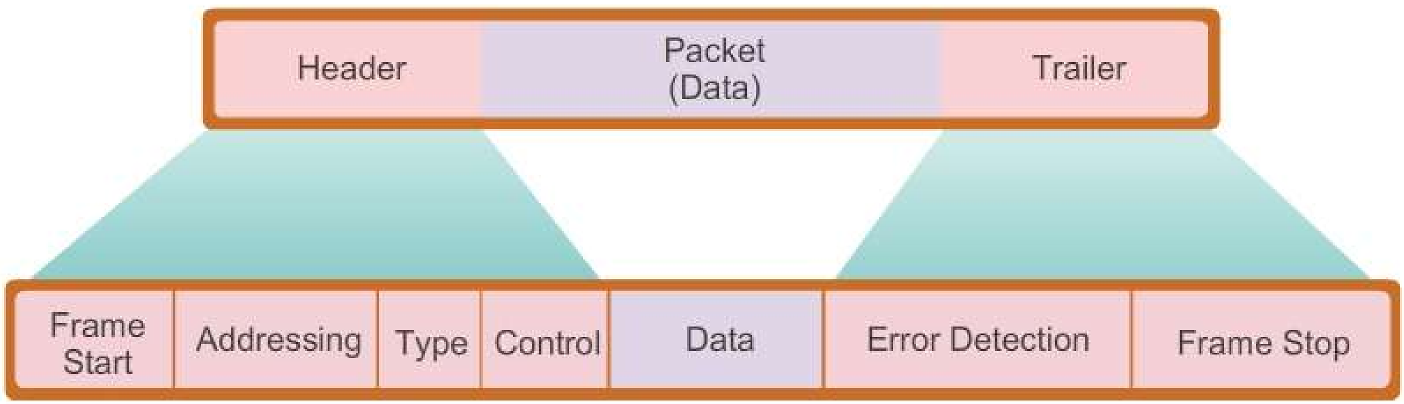
- Different media access control methods may be required during the course of a single communication. Each network environment that packets encounter as they travel from a local host to a remote host can have different characteristics.

Data Link Layer



Data Link Layer

- Data link layer frame includes:
  - **Header**: Contains control information, such as addressing, and is located at the beginning of the PDU.
  - **Data**: Contains the IP header, transport layer header, and application data.
  - **Trailer**: Contains control information for error detection added to the end of the PDU.



Data Link Layer

Ethernet Protocol

A Common Data Link Layer Protocol for LANs

	Frame					
Field name	Preamble	Destination	Source	Type	Data	Frame Check Sequence
Size	8 bytes	6 bytes	6 bytes	2 bytes	46 - 1500 bytes	4 bytes

Preamble - Used for synchronization; also contains a delimiter to mark the end of the timing information
Destination Address - 48-bit MAC address for the destination node
Source Address - 48-bit MAC address for the source node
Type - Value to indicate which upper layer protocol will receive the data after the Ethernet process is complete
Data or payload - This is the PDU, typically an IPv4 packet, that is to be transported over the media.
Frame Check Sequence (FCS) - A value used to check for damaged frames

Data Link Layer

Point-to-Point Protocol

A Common Data Link Protocol for WANs

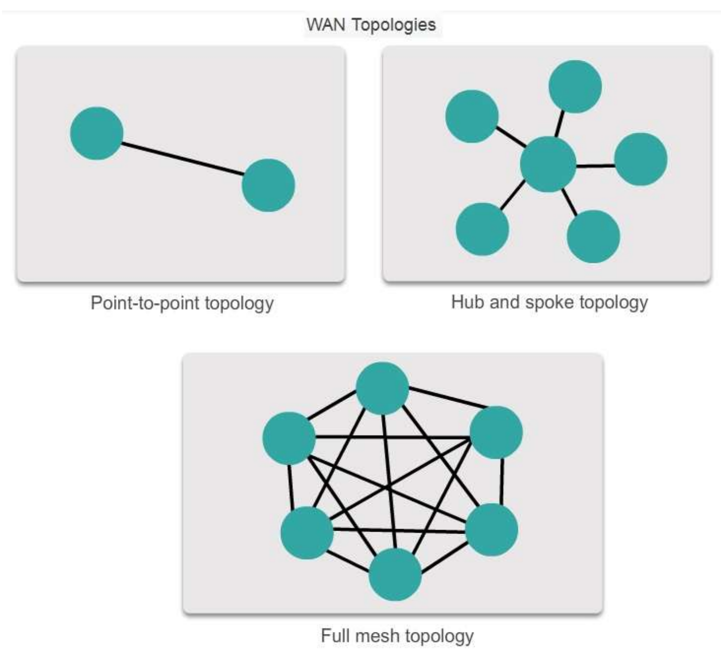
Frame

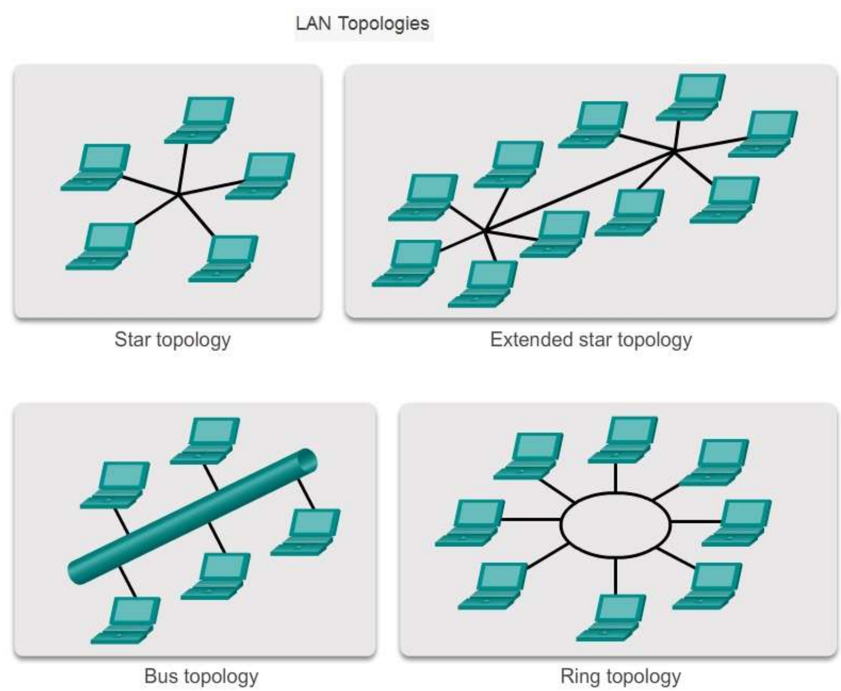
Field name	Flag	Address	Control	Protocol	Data	FCS
Size	1 byte	1 byte	1 byte	2 bytes	variable	2 or 4 bytes

Flag - A single byte that indicates the beginning or end of a frame. The flag field consists of the binary sequence 01111110.
Address - A single byte that contains the standard PPP broadcast address. PPP does not assign individual station addresses.
Control - A single byte that contains the binary sequence 00000011, which calls for transmission of user data in an unsequenced frame.
Protocol - Two bytes that identify the protocol encapsulated in the data field of the frame. The most up-to-date values of the protocol field are specified in the most recent Assigned Numbers Request For Comments (RFC).
Data - Zero or more bytes that contain the datagram for the protocol specified in the protocol field.
Frame Check Sequence (**FCS**) - Normally 16 bits (2 bytes). By prior agreement, consenting PPP implementations can use a 32-bit (4-byte) FCS for improved error detection.

Media Access Control

- There are different methods of controlling access to the media. These methods depends on:
  - **Topology**: How the connection between the nodes appears to the data link layer.
  - **Media sharing**: How the nodes share the media.



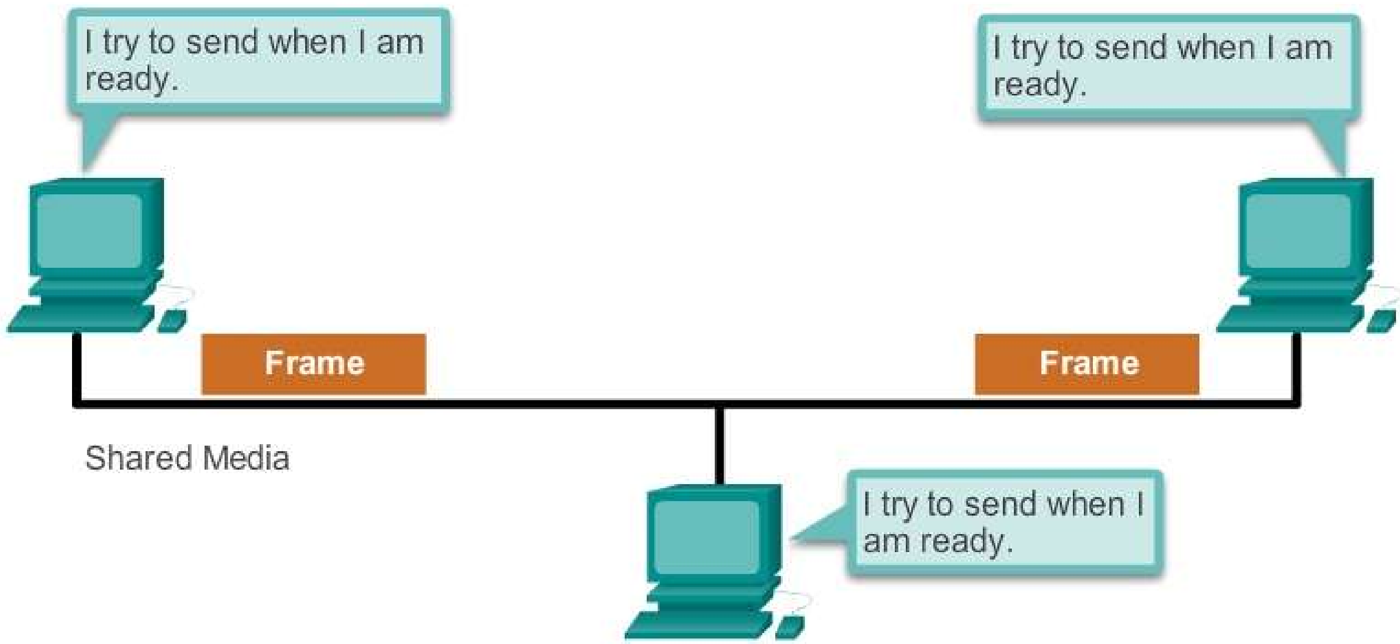


Media Access Control

- There are two basic media access control methods for shared media:
- **Contention-based access**: All nodes compete for the use of the medium but have a plan if there are collisions.
- **Controlled access**: Each node has its own time to use the medium.
- The contention-based systems do not scale well under heavy media use. As the number of nodes increases, the probability of collision increases.
- CSMA is usually implemented in conjunction with a method for resolving the media contention.

Media Access Control

Contention-Based Access

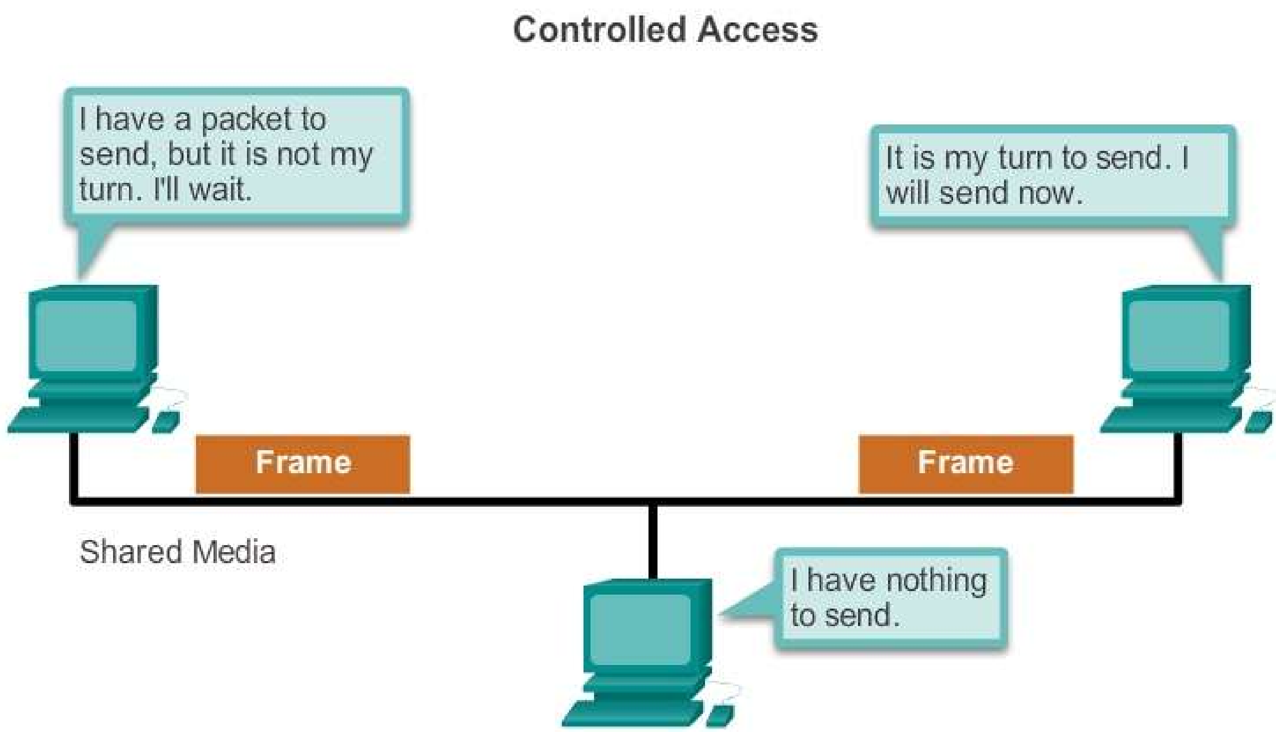


Characteristics	Contention-Based Technologies
• Stations can transmit at any time • Collisions exist • There are mechanisms to resolve contention for the media	• CSMA/CD for 802.3 Ethernet networks • CSMA/CA for 802.11 wireless networks

Media Access Control

- **Carrier sense multiple access with collision detection (CSMA/CD)**: The end device monitors the media for the presence of a data signal. If a data signal is absent and therefore the media is free, the device transmits the data. If signals are then detected that show another device was transmitting at the same time, all devices stop sending and try again later.
- **Carrier sense multiple access with collision avoidance (CSMA/CA)**: The end device examines the media for the presence of a data signal. If the media is free, the device sends a notification across the media of its intent to use it. Once it receives a clearance to transmit, the device then sends the data.

Media Access Control



Characteristics	Controlled Access Technologies
• Only one station transmits at a time • Devices wishing to transmit must wait their turn • No collisions • May use a token passing method	• Token Ring (IEEE 802.5) • Fiber Distributed Data Interface (FDDI)

Ethernet

- Ethernet is the most widely used LAN technology used today. Ethernet operates in the data link layer and the physical layer. It is a family of networking technologies that are defined in the IEEE 802.2 and 802.3 standards.
- Ethernet supports data bandwidths of: 10Mbps, 100Mbps, 1Gbps, 10Gbps, 40Gbps, 100Gbps.
- The minimum Ethernet frame size is **64 bytes** and the maximum is **1518 bytes**.

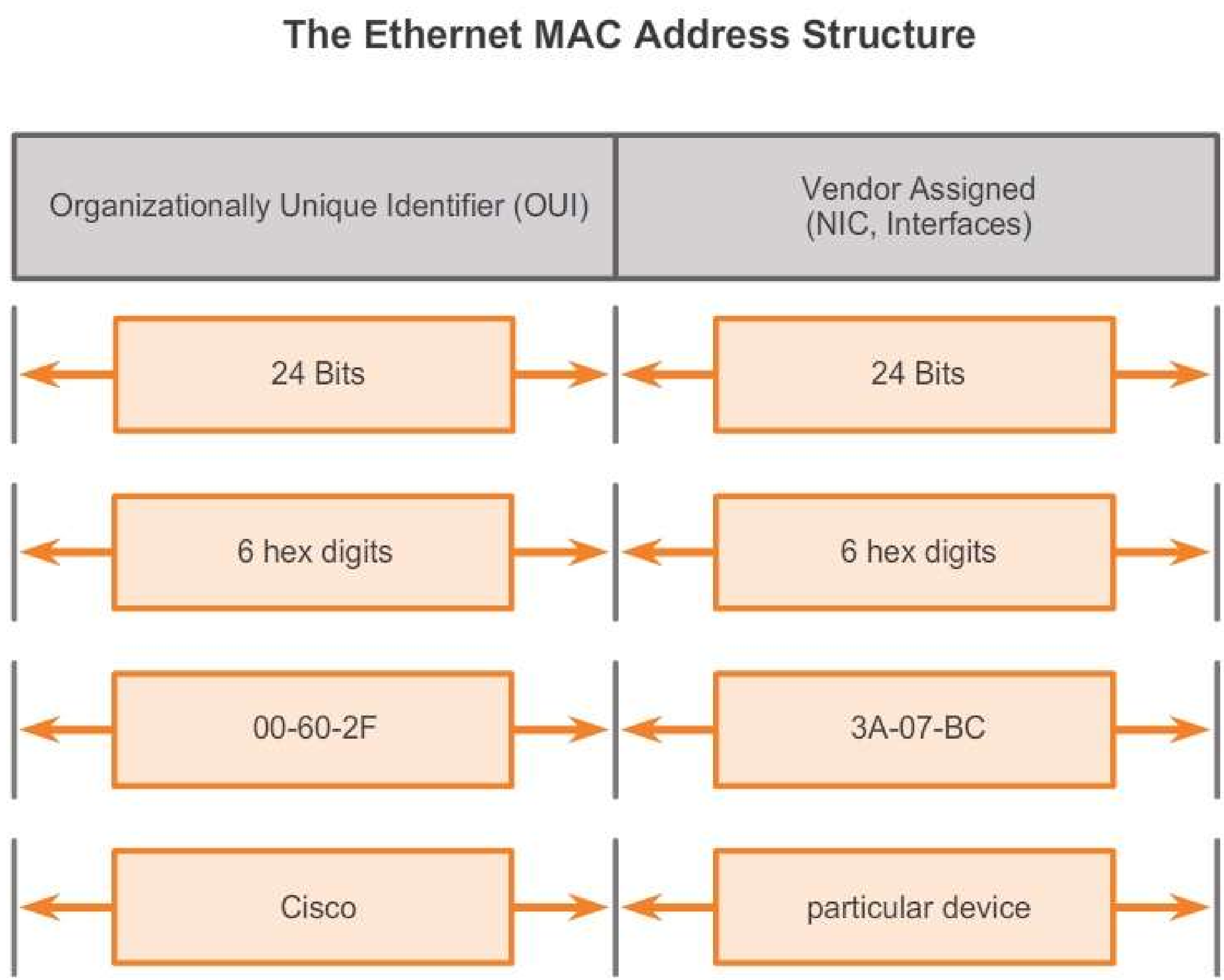
Ethernet

Field	Description
Preamble and Start Frame Delimiter Fields	The Preamble (7 bytes) and Start Frame Delimiter (SFD), also called the Start of Frame (1 byte), fields are used for synchronization between the sending and receiving devices. These first eight bytes of the frame are used to get the attention of the receiving nodes. Essentially, the first few bytes tell the receivers to get ready to receive a new frame.
Destination MAC Address Field	This 6-byte field is the identifier for the intended recipient. As you will recall, this address is used by Layer 2 to assist devices in determining if a frame is addressed to them. The address in the frame is compared to the MAC address in the device. If there is a match, the device accepts the frame. Can be a unicast, multicast or broadcast address.
Source MAC Address Field	This 6-byte field identifies the originating NIC or interface of the frame.
Type / Length	This 2-byte field identifies the upper layer protocol encapsulated in the Ethernet frame. Common values are, in hexadecimal, 0x800 for IPv4, 0x86DD for IPv6 and 0x806 for ARP. **Note**: You may also see this field referred to as EtherType, Type, or Length.
Data Field	This field (46 - 1500 bytes) contains the encapsulated data from a higher layer, which is a generic Layer 3 PDU, or more commonly, an IPv4 packet. All frames must be at least 64 bytes long. If a small packet is encapsulated, additional bits called a pad are used to increase the size of the frame to this minimum size.
Frame Check Sequence Field	The Frame Check Sequence (FCS) field (4 bytes) is used to detect errors in a frame. It uses a cyclic redundancy check (CRC). The sending device includes the results of a CRC in the FCS field of the frame. The receiving device receives the frame and generates a CRC to look for errors. If the calculations match, no error occurred. Calculations that do not match are an indication that the data has changed; therefore, the frame is dropped. A change in the data could be the result of a disruption of the electrical signals that represent the bits.

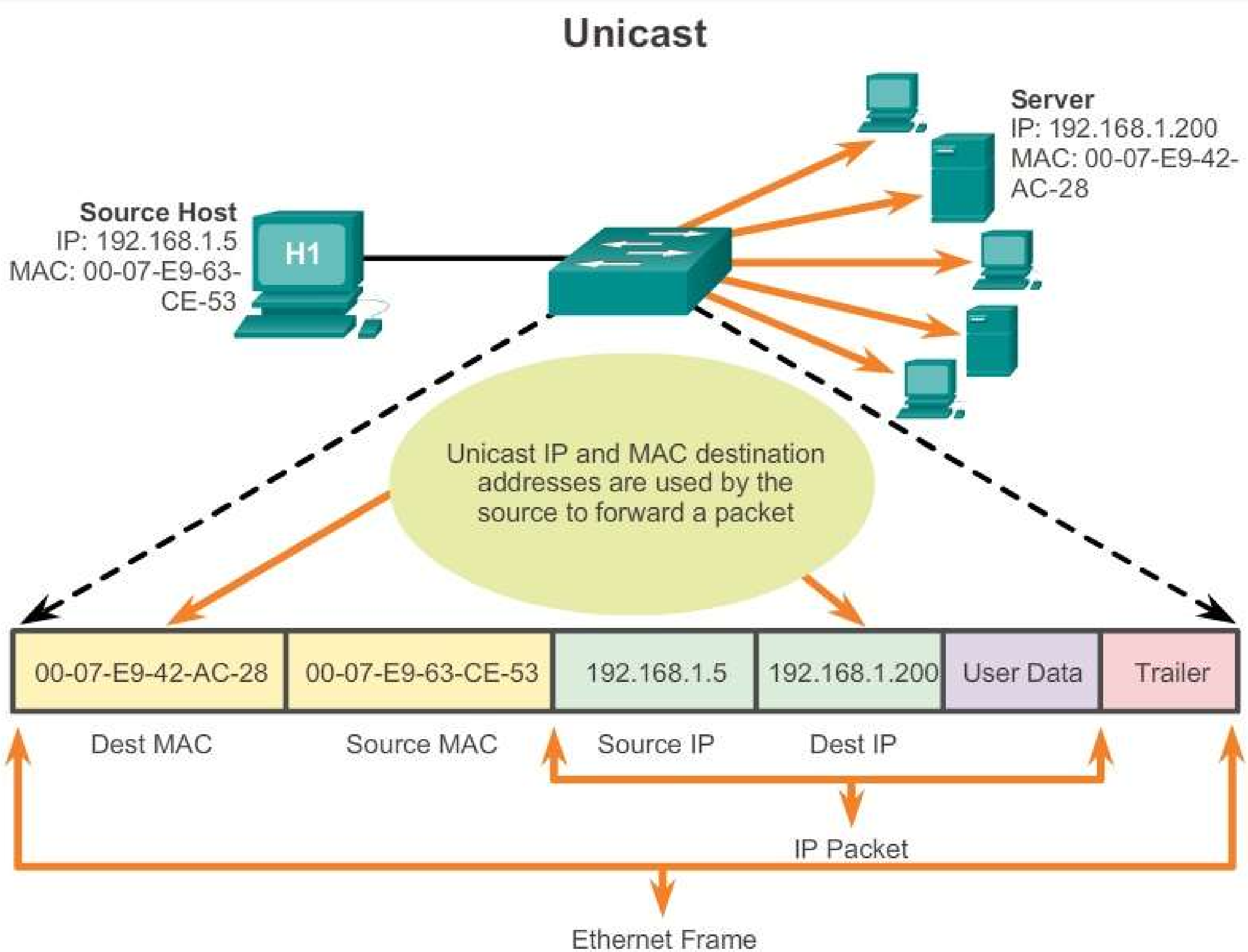
Ethernet

- A unique identifier called a **MAC address** was created to identify the actual source and destination nodes within an Ethernet network. An Ethernet MAC address is a **48-bit** binary value expressed as **12 hexadecimal digits** (4 bits per hexadecimal digit).
- In Ethernet, different MAC addresses are used for Layer 2 unicast, broadcast, and multicast communications.
- **ipconfig /all** command can be used to identify the MAC address of an Ethernet adapter.

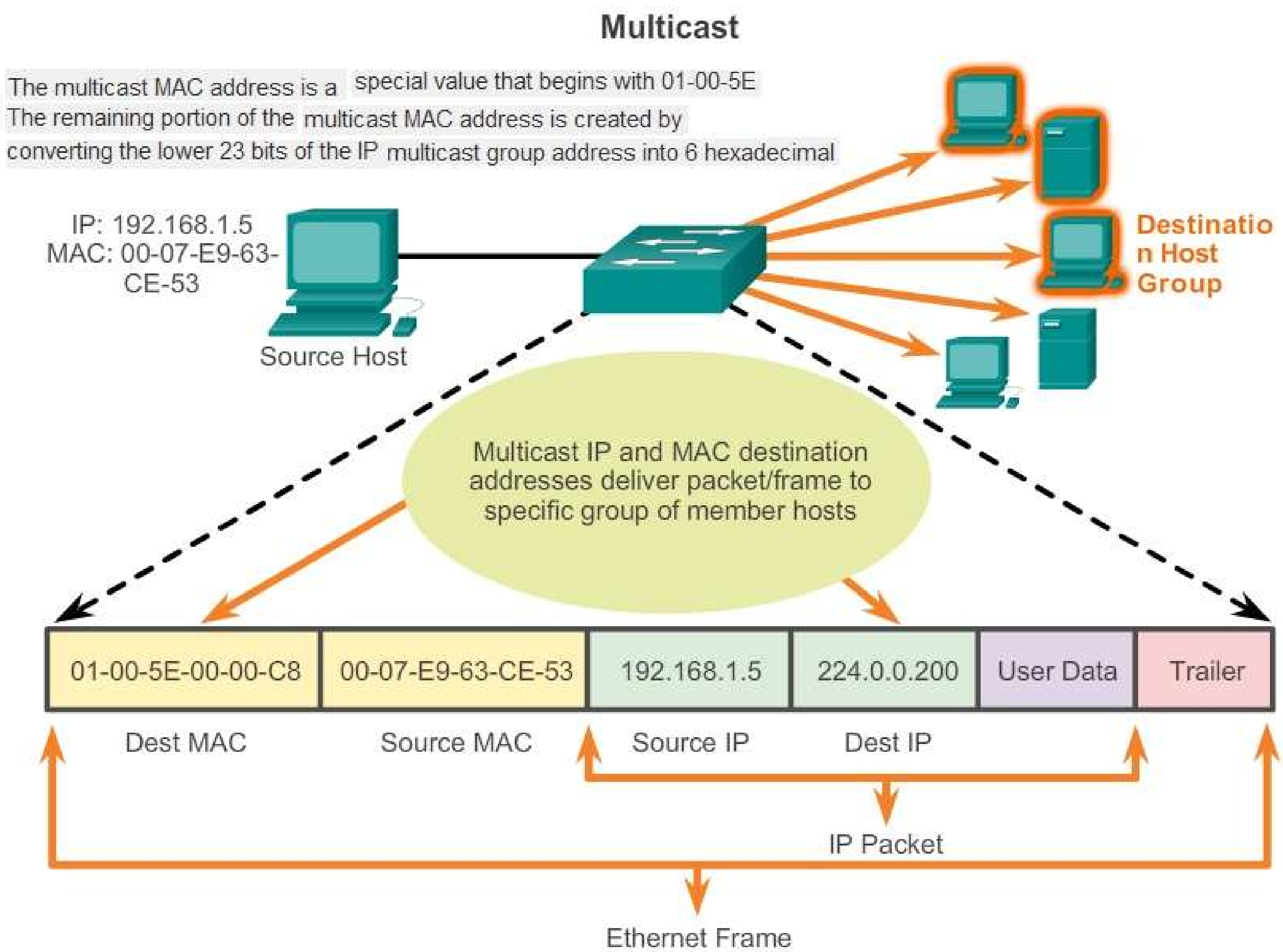
Ethernet


The Ethernet MAC Address Structure
Organizationally Unique Identifier (OUI)
Vendor Assigned
(NIC, Interfaces)
24 Bits
24 Bits
6 hex digits
6 hex digits
00-60-2F
3A-07-BC
Cisco
particular device

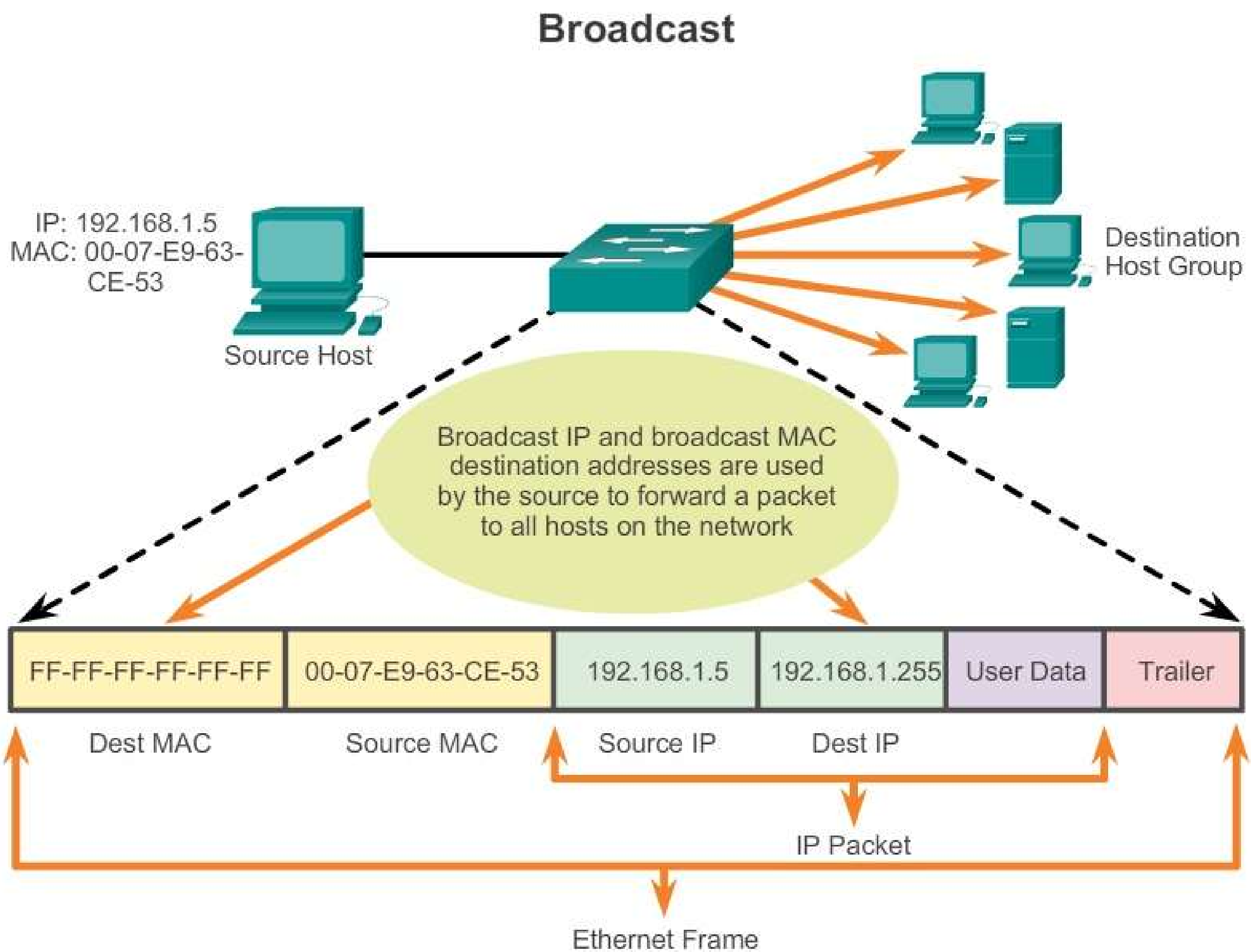
Ethernet


Unicast
Server
IP: 192.168.1.200
MAC: 00-07-E9-42-AC-28
Source Host
IP: 192.168.1.5
MAC: 00-07-E9-63-CE-53
H1
Unicast IP and MAC destination addresses are used by the source to forward a packet
00-07-E9-42-AC-28
00-07-E9-63-CE-53
192.168.1.5
192.168.1.200
User Data
Trailer
Dest MAC
Source MAC
Source IP
Dest IP
IP Packet
Ethernet Frame

Ethernet


Multicast
The multicast MAC address is a special value that begins with 01-00-5E
The remaining portion of the multicast MAC address is created by
converting the lower 23 bits of the IP multicast group address into 6 hexadecimal
IP: 192.168.1.5
MAC: 00-07-E9-63-CE-53
Source Host
Destination Host Group
Multicast IP and MAC destination addresses deliver packet/frame to specific group of member hosts
01-00-5E-00-00-C8
00-07-E9-63-CE-53
192.168.1.5
224.0.0.200
User Data
Trailer
Dest MAC
Source MAC
Source IP
Dest IP
IP Packet
Ethernet Frame

Ethernet

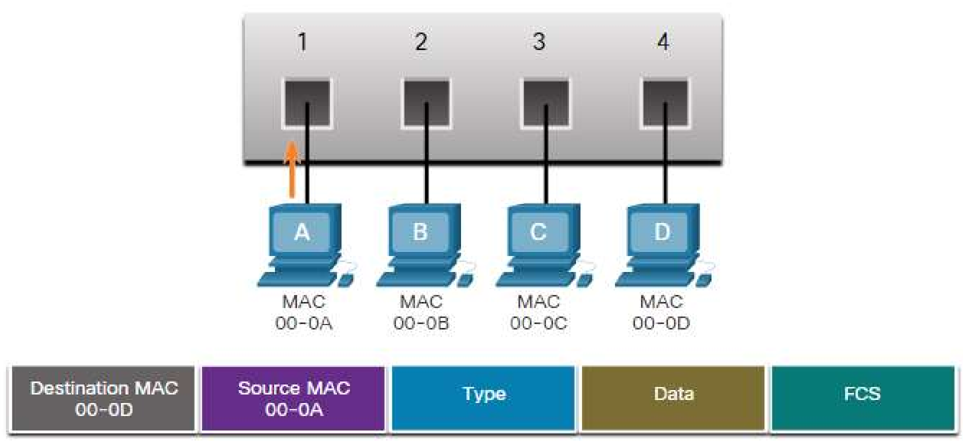


Broadcast
IP: 192.168.1.5
MAC: 00-07-E9-63-CE-53
Source Host
Destination Host Group
Broadcast IP and broadcast MAC destination addresses are used by the source to forward a packet to all hosts on the network
FF-FF-FF-FF-FF-FF
00-07-E9-63-CE-53
192.168.1.5
192.168.1.255
User Data
Trailer
Dest MAC
Source MAC
Source IP
Dest IP
IP Packet
Ethernet Frame

Ethernet

- Ethernet switch examines its MAC address table to make a forwarding decision for each frame.
- The switch dynamically builds the MAC address table by examining the **source MAC address** of the frames received on a port.

MAC Address Table	
Port	**MAC Address**
1	00-0A



Switching

- Switches used one of the following forwarding methods for switching data between network ports:
  - In **store-and-forward switching**, when the switch receives the frame, it stores the data in buffers until the **complete frame** has been received. During the storage process, the switch performs an error check using the Cyclic Redundancy Check (CRC)
  - In **cut-through switching**, the switch acts upon the data as soon as it is received, even if the transmission is not complete. There are two variants of cut-through switching:
    - **Fast-forward switching**: Fast-forward switching immediately forwards a packet after reading the destination address.

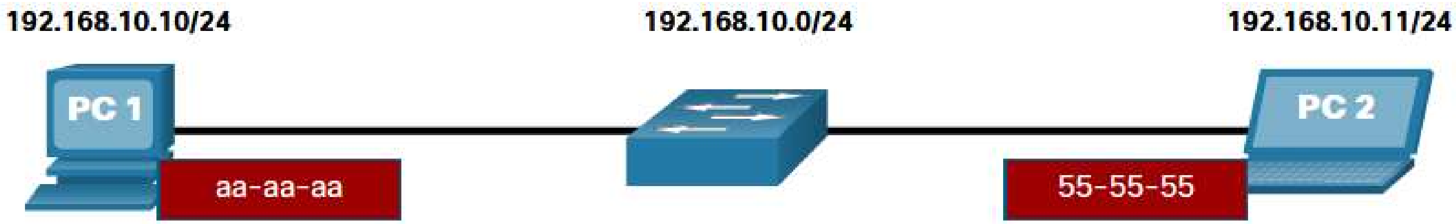
Switching

- **Fragment-free switching**: In fragment-free switching, the switch stores the **first 64 bytes** of the frame before forwarding.

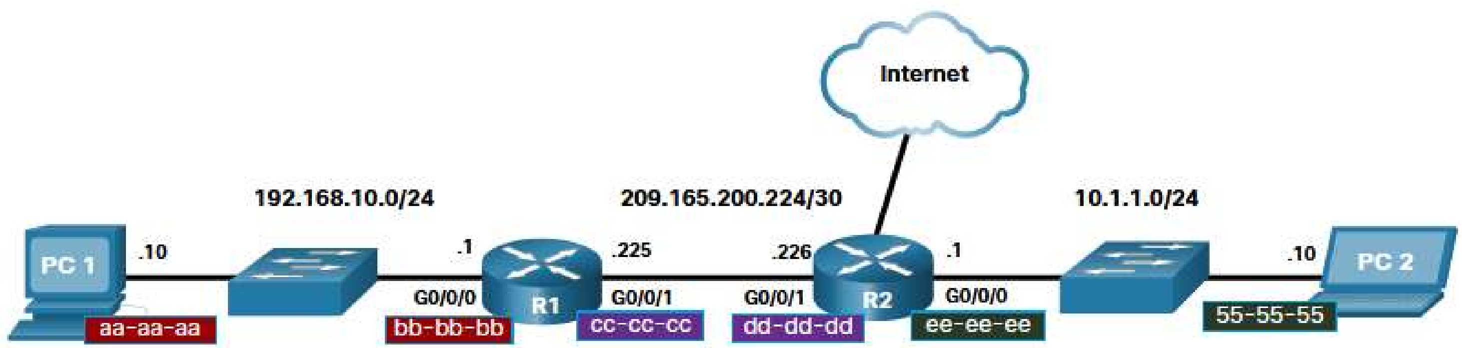
- An Ethernet switch may use a buffering technique to store frames before forwarding them.
- Buffering may also be used when the destination port is busy because of congestion. The switch stores the frame until it can be transmitted.

Port-based memory	• Frames are stored in queues that are linked to specific incoming and outgoing ports. • A frame is transmitted to the outgoing port only when all the frames ahead in the queue have been successfully transmitted. • It is possible for a single frame to delay the transmission of all the frames in memory because of a busy destination port. • This delay occurs even if the other frames could be transmitted to open destination ports.
Shared memory	• Deposits all frames into a common memory buffer shared by all switch ports and the amount of buffer memory required by a port is dynamically allocated. • The frames in the buffer are dynamically linked to the destination port enabling a packet to be received on one port and then transmitted on another port, without moving it to a different queue.

Address Resolution Protocol

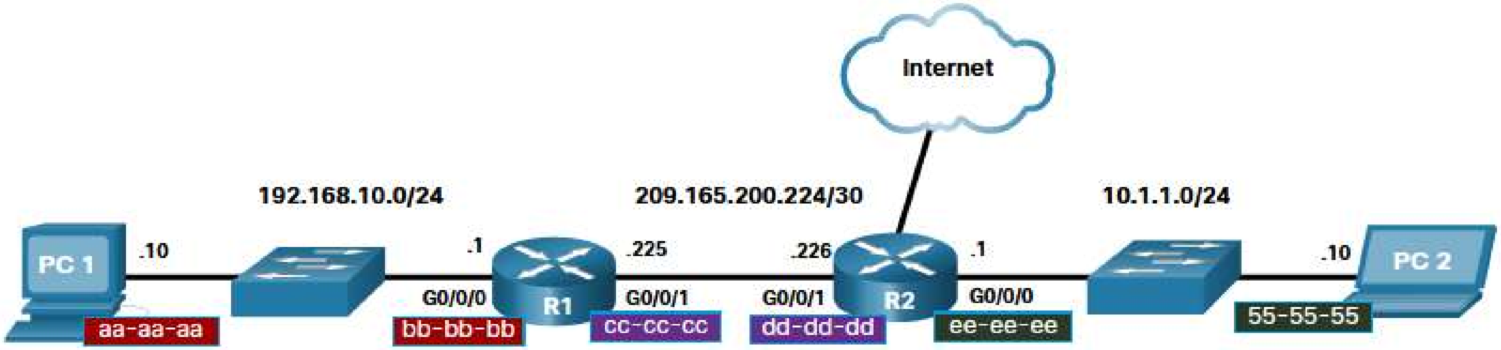


Destination MAC	Source MAC	Source IPv4	Destination IPv4
55-55-55	aa-aa-aa	192.168.10.10	192.168.10.11

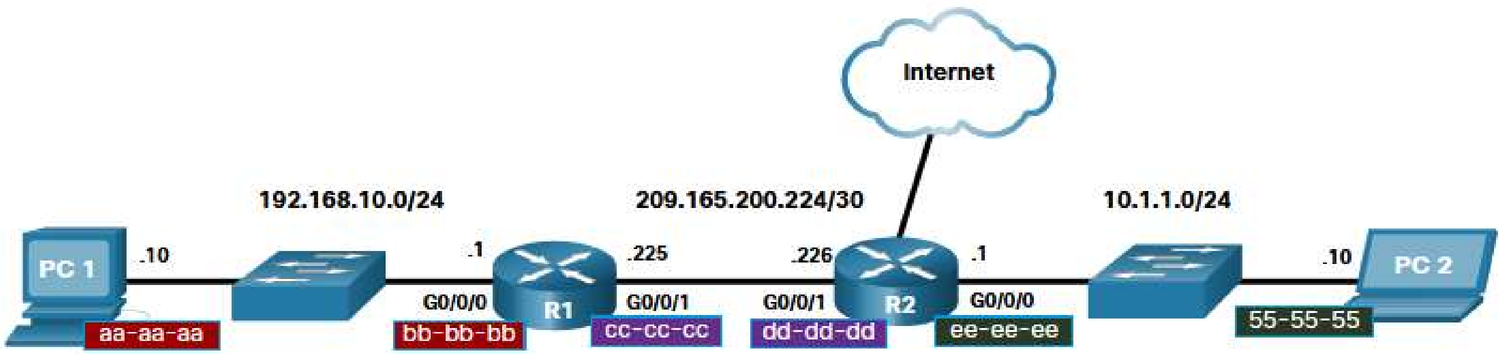


Destination MAC	Source MAC	Source IPv4	Destination IPv4
bb-bb-bb	aa-aa-aa	192.168.10.10	10.1.1.10

Address Resolution Protocol



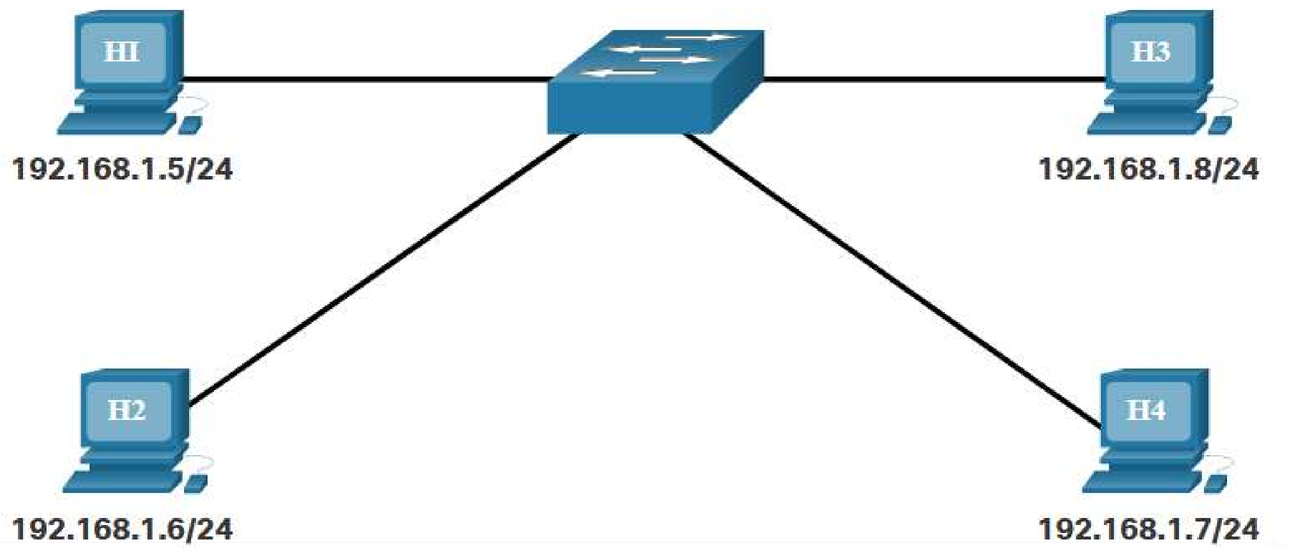
Destination MAC	Source MAC	Source IPv4	Destination IPv4
dd-dd-dd	cc-cc-cc	192.168.10.10	10.1.1.10



Destination MAC	Source MAC	Source IPv4	Destination IPv4
55-55-55	ee-ee-ee	192.168.10.10	10.1.1.10

Address Resolution Protocol

I need to send information to 192.168.1.7, but I only have the IP address. I don't know the MAC address of the device that has that IP.



Address Resolution Protocol

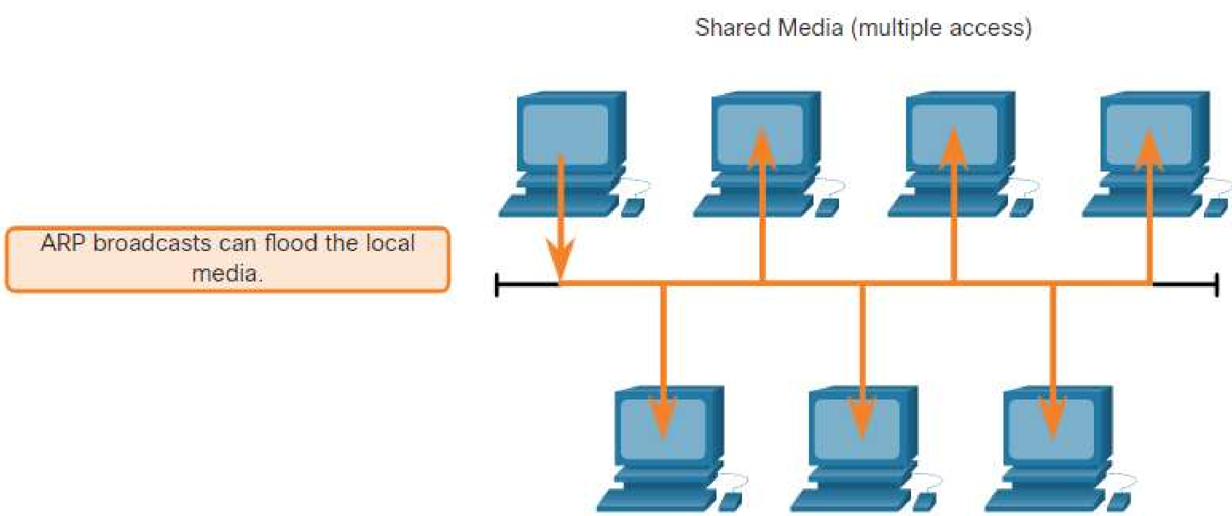
- ARP is used to **resolve IPv4 Addresses to MAC Addresses**.
- To begin the process, a transmitting node attempts to locate the MAC address mapped to an IPv4 destination. If this map is found in the table, the node uses the MAC address as the destination MAC in the frame that encapsulates the IPv4 packet.
- If an entry is not found, the ARP processes then send out an **ARP request packet (FF-FF-FF-FF-FF-FF)** to discover the MAC address of the destination device on the local network. If a device receiving the request has the destination IP address, it responds with an ARP reply. A map is created in the ARP table.

Address Resolution Protocol

- If the destination IPv4 address is on a different network than the source IPv4 address, the device will search the ARP table for the IPv4 address of the **default gateway**.
- If no device responds to the ARP request, the packet is dropped because a frame cannot be created.
- For each device, an ARP cache timer removes ARP entries that have not been used for a specified period of time. It is between 15 and 45 seconds. (Windows)
- ARP Issues
  - Overhead on the Media
  - Security

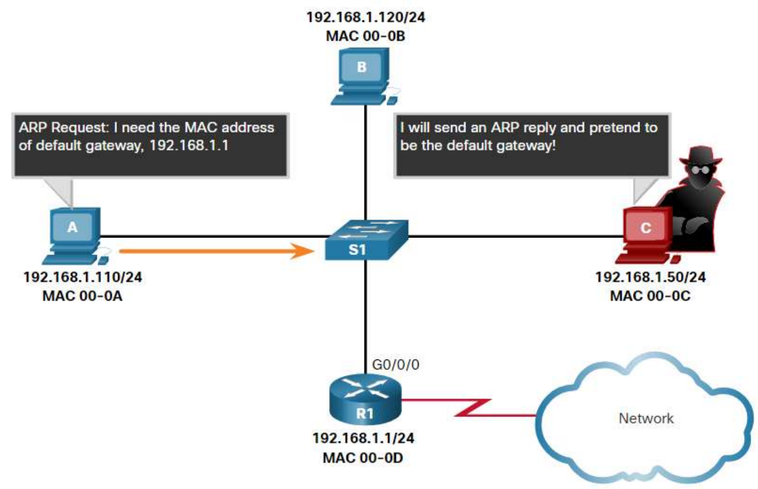
Address Resolution Protocol

- As a broadcast frame, an ARP request is received and processed by every device on the local network. If a large number of devices start accessing network services at the same time, there could be some reduction in performance for a short period of time



Address Resolution Protocol

- A threat actor can use ARP spoofing to perform an ARP poisoning attack. This is a technique used by a threat actor to reply to an ARP request for an IPv4 address belonging to another device, such as the default gateway
- The threat actor sends an ARP reply with its own MAC address. The receiver of the ARP reply will add the wrong MAC address to its ARP table and send these packets to the threat actor.



Address Resolution Protocol

```
Router#show ip arp

                              Age
Protocol   Address           (min)  Hardware Addr    Type   Interface
Internet   172.16.233.229     -     0000.0c59.f892   ARPA   Ethernet0/0
Internet   172.16.233.218     -     0000.0c07.ac00   ARPA   Ethernet0/0
Internet   172.16.168.11      -     0000.0c63.1300   ARPA   Ethernet0/0
Internet   172.16.168.254     9     0000.0c36.6965   ARPA   Ethernet0/0
```

```
C:\>arp -a

Interface: 192.168.1.67 --- 0xa
  Internet Address      Physical Address      Type
  192.168.1.254         64-0f-29-0d-36-91     dynamic
  192.168.1.255         ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static
```